Arun Paudel

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Education

Simon Fraser University — *Burnaby, BC*

Bachelor of Applied Sciences in Computer Science, Minor in Statistics — **Expected Graduation: December 2027**

- **Leadership:** Area Coordinator, SFU Residence and Housing; team of **9 Resident Advisors**.

Capstone Research

PhishX & PhishForge | *Python / SPARQL / ChromaDB / Ollama / HPC / Apptainer* — **Jan 2026 to Present**

- Translated published phishing models into deployable **Python** pipelines under faculty supervision.
- Optimized model inference by **2.2**× while maintaining **97.85% F1** through benchmarking.
- Accelerated inference with collaborators by **37**× (**48s to 1.3s/URL**) via containers and model warmup.
- Created an extensible evaluation harness to benchmark generated datasets and document results for researchers.
- Devised a **generative AI** pipeline to produce benchmark data and document methods for fellow researchers.

Technical Experience

Backend Developer — *Burnaby, BC*

SFU Computer Science Student Society — **Apr 2026 to Present**

- Shipped a **FastAPI** and **PostgreSQL** events service for kiosks displaying upcoming campus events.
- Refined implementations through **GitHub** pull request reviews with a senior developer to meet team standards.
- Structured event schemas and reversible **Alembic** migrations in **PostgreSQL** to support safe rollbacks.

Projects

PokéDrop: Reliable Card Reservations | *TypeScript / React / Go / AWS* — **GitHub**

- Architected a **Go** reservation API with **Lambda**, **API Gateway**, and **SQS** to queue card claims and decouple request processing.
- Orchestrated atomic **DynamoDB** transactions and idempotency checks to prevent overselling and duplicate deductions on retries.
- Delivered a **React/TypeScript** dashboard with **Cognito** authentication, card inventory views, and status updates for reservations.
- Established repeatable deployments with **AWS CDK** and **GitHub Actions**, provisioning resources with scoped **IAM** permissions.
- Directed duplicate request and worker failure tests, configured **CloudWatch** alarms, and documented recovery for failed messages.

Clash of Typers | *Python / TCP / Concurrency / Protocol Design* — **GitHub**

- Engineered a **Python TCP** server with **select** I/O multiplexing to synchronize **25 shared locks** and scores across game clients.
- Defined handlers for **5 JSON request types**, using newline framing and per client buffers to reconstruct partial **TCP** messages.
- Enforced exclusive lock claims and server validation of typing accuracy and speed to reject conflicting actions and award points.

Fort Defense Game | *Java / Spring Boot / MongoDB / JUnit / Swagger* — **GitHub**

- Designed **9 REST API routes** with **Spring Boot** for games and users, mapping invalid input and missing IDs to **HTTP** errors.
- Developed **59 JUnit tests** covering board state, coordinate validation, scoring, and DTO mapping to guard against regressions.
- Standardized **Swagger** documentation for **6 API operations**, specifying parameters and response codes for frontend integration.

Web Generator | *Python / Object Oriented Programming / CLI Tools / GitHub Pages* — **GitHub**

- Constructed a **Python** site generator with **6 Markdown block types**, **zero external dependencies**, and recursive rendering.
- Implemented **68 unit tests** with **unittest** to validate parsing, inline formatting, HTML rendering, and malformed input handling.
- Automated **GitHub Pages** builds with shell scripts and configurable base paths, rewriting links and asset URLs for project hosting.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL, TypeScript, Go (Golang), HTML, CSS, SPARQL
Frameworks: React, Vite, FastAPI, Flask, Spring Boot, REST APIs
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, ChromaDB, Alembic migrations
AWS: Lambda, API Gateway, SQS, Cognito, S3, CloudFront, IAM, CloudWatch, CDK
Systems: TCP/IP, JSON protocols, I/O multiplexing, concurrency, locking, atomic transactions, idempotency, retries
Tools & Delivery: Git, GitHub Actions, CI/CD, infrastructure as code, GitHub Pages, shell scripting, CLI tools, Linux, Docker, Apptainer, HPC
Testing & Operations: JUnit, unittest, Go testing, Vitest, k6, Swagger, Postman, OpenAPI, monitoring, queue recovery
AI & Engineering: Ollama, generative AI, benchmarking, data structures and algorithms, object oriented programming, SDLC, Agile, code review